Smart campus crowd management systems

"How can real-time crowd monitoring, predictive analytics, and intelligent route optimization be integrated to detect emerging congestion and dynamically guide pedestrian movement in university campuses? Examine sensor-based occupancy estimation, privacy-preserving monitoring, alternative route recommendations, automated alerts, and AI-driven crowd management, identifying research gaps and opportunities for scalable smart-campus solutions." *Generated by LeapSpace*

Smart-Campus Crowd Monitoring and Dynamic Pedestrian Guidance at a Glance

Key Findings Table

Topic	Key finding	Campus relevance	Citations
Occupancy sensing	Cameras and LiDAR provide the strongest fine-grained counting accuracy at chokepoints, while Wi-Fi/CSI and BLE scale coverage but are noisier under dense, mixed indoor-outdoor conditions [1] [2] [3].	Best for entrances, corridors, plazas, sidewalks, and high-flow transitions [2] [3].	[1] [2] [3] [4] [5]
Sensor fusion	Fusion improves robustness, but the evidence base is fragmented and lacks head-to-head campus benchmarks across camera, Wi-Fi, BLE, LiDAR, and fusion in the same setting [2] [6].	Supports campus-wide mixed-modality deployment, but benchmarking remains a major gap [6] [7].	[2] [6] [7]
Predictive analytics	Short-horizon early warning from anonymized Wi-Fi data is feasible; one campus canteen study detected an early peak before the rush with useful agreement and coverage [8].	Suitable for class-transition surges, dining peaks, and corridor buildup [8] [9].	[8] [9] [10]

Congestion thresholds	Density and congestion thresholds exist, but calibration for live university operations is weak and not standardized across space types [11] [12] [13].	Campuses need local tuning for canteens, lecture halls, libraries, and outdoor walkways [9] [14].	[9] [11] [12] [13] [14]
Privacy-preserving monitoring	Monitoring is shifting toward edge inference, metadata-only export, anonymized wireless features, rotating identifiers, and federated learning [15] [16] [17].	Compatible with continuous campus monitoring while reducing raw personal-data retention [18] [19].	[15] [16] [17] [18] [19]
Route recommendation	Multi-criteria campus routing can encode congestion, stairs, slope, elevators, covered paths, and user preferences, but direct evidence on live reroute adherence and time savings is minimal [20] [21] [22].	Strong design basis for inclusive guidance, weak field evidence on outcomes [23] [24].	[20] [21] [22] [23] [24]
Alerts and operations	Automated alerts, dashboards, and role-based access are common, but override procedures, repeat-notification logic, and event-level crowd-control metrics are thinly documented [14] [25] [26].	Suggests campuses need human-in-the-loop command workflows, not just detection models [27] [28].	[14] [25] [26] [27] [28]
Scalability	Layered edge-cloud, federated, and standards-oriented architectures are repeatedly proposed, but most evidence remains single-building or short-duration [29] [30] [31].	Scalable smart-campus deployment is plausible but not yet mature [32] [33].	[29] [30] [31] [32] [33]

Direct Answer

An effective university-campus system should integrate five layers into a closed loop: **multimodal sensing**, **short-horizon prediction**, **thresholded alerting**, **dynamic route guidance**, and **human-supervised operations** [2] [3] [8] [10] [25] [34]. The strongest practical design is an **edge-first "sentinel chokepoint + ambient mesh" architecture**: use cameras or LiDAR at entrances, stairs, bottlenecks, and outdoor sidewalks for accurate counts, and use Wi-Fi/CSI or BLE for broader room-, floor-, and walkway-level occupancy fields where full visual coverage is impractical [1] [3] [5] [35]. These streams should feed a zone graph that forecasts occupancy buildup 5-15 minutes ahead using AP counts, trajectory features, or graph-based models, then triggers advisories when density, occupancy-change, or dwell-time thresholds are exceeded [8] [10] [11] [36].

For privacy, campuses should process video and wireless data locally, export only counts or anonymized features, rotate identifiers, and consider federated learning for cross-building model updates; however, privacy-enhancing methods create utility and resource tradeoffs that must be explicitly managed [15] [16] [37] [38] [39]. For guidance, route costs should be dynamically reweighted by predicted congestion, accessibility constraints, weather/shelter, and operational restrictions, then communicated via mobile apps, digital signage, or dashboards [21] [34] [40]. The main gap is not sensing or prediction alone, but **field-validated closed-loop actuation**: the supplied evidence does not show a mature campus deployment that detects congestion, predicts near-future buildup, reroutes pedestrians, and then measures adherence, delay reduction, and safety improvement end-to-end [8] [22] [24] [41].

Confidence level: High - the evidence consistently supports the integrated architecture, but **Medium** on quantified operational benefits because closed-loop campus evaluations are sparse [2] [8] [15] [29] [34].

Study Scope

This report synthesizes evidence on university-campus and closely related smart-facility systems spanning occupancy sensing, congestion prediction, privacy-preserving monitoring, dynamic routing, alerts, and deployment architectures from approximately 2018-2025, using empirical campus studies, applied system papers, and adjacent operational crowd-management work where campus-specific evidence is thin [7] [10] [15] [29] [42].

Assumptions & Limitations

The evidence base is strongest for sensing and prediction, weaker for closed-loop rerouting and operator workflow outcomes; many deployments are single-building, single-modality, or short-duration, and fairness analysis across campus groups is largely absent [6] [9] [22] [32] [43]. Threshold values reported in the literature are rarely calibrated to live campus contexts across canteens, lecture halls, libraries, and walkways, so they should be treated as starting points rather than universal rules [11] [13] [14].

Suggested Further Research

- Run a **head-to-head campus benchmark** comparing camera, Wi-Fi/CSI, BLE, LiDAR, and fusion under the same dense indoor-outdoor conditions [6] [7].
- Evaluate **closed-loop rerouting** with adherence, travel-time savings, bottleneck reduction, and perceived fairness as primary outcomes [14] [22] [24].
- Standardize **campus alert metrics** such as warning lead time, false alarms per hour, missed-event rate, and operator response time [14] [44].
- Test **edge-vs-cloud tradeoffs** for latency, privacy, bandwidth, and model utility in multi-building deployments [15] [31] [45].
- Add **fairness and accessibility audits** across students, staff, visitors, and users with disabilities [43] [46] [47].

Full Report Body

Foundational Context

Campus crowd management is an integrated control problem rather than a single sensing task. At minimum, it requires: **(1) occupancy estimation** to know how many people are where, **(2) forecasting** to anticipate whether a local buildup will become congestion, **(3) intervention logic** to decide when to alert or reroute, and **(4) communication channels** that can influence pedestrian movement through apps, signage, or operator actions [8] [11] [25] [34]. The literature shows that these elements are often studied separately; sensing is relatively mature, forecasting is improving, but the detect-predict-guide loop is only partially demonstrated in campus settings [2] [10] [22] [41].

A useful technical framing is to treat the campus as a **spatiotemporal graph** of buildings, corridors, entrances, plazas, sidewalks, transit stops, and vertical connectors. Sensors estimate occupancy and flow on nodes and edges; prediction models estimate near-future state changes; route optimizers reweight path costs according to expected congestion and accessibility constraints; operators supervise the process through dashboards and alerts [10] [34] [48]. Taken together, these findings suggest that a scalable campus system should be **zone-based rather than person-tracking-based**, because many privacy-preserving sensing pipelines already work by aggregated counts, AP-level telemetry, or region-level density rather than exact identity-linked trajectories [16] [49] [50].

Confidence level: High - this framing is directly supported across sensing, privacy, routing, and deployment studies [8] [15] [29] [34].

1. Real-Time Occupancy Sensing and Congestion Prediction

1.1 Sensor-based occupancy estimation

The evidence shows a consistent tradeoff: **visual systems are more accurate locally; wireless systems cover more space at lower cost; fusion improves robustness** [1] [2] [3] [4] [5]. Camera-based counting is repeatedly treated as the high-accuracy reference, with reports above 95% accuracy against manual counts in campus/building settings, although performance can degrade under low light, occlusion, and high density [1] [51]. Multi-camera fusion further improves crowded-entrance counting, with one fused two-camera system reaching about 98.1% accuracy and outperforming prior methods [2]. For dense outdoor sidewalks, 2D LiDAR also performs strongly, with reported disaggregate counting accuracy of 97.4% and 98.7% on two sidewalks near a university environment [3].

Wi-Fi methods are more mixed. Institutional and academic-building studies report around 87% average accuracy in some settings, but also note that Wi-Fi is not good for precise zone-level localization and can count people outside a building as inside because of coarse spatial resolution [1] [4] [52]. RSSI/MAC-based occupancy detection can perform very well in some experiments, reaching 98.85% accuracy with low normalized count error, while CSI-based inference reached 90% accuracy in academic settings [5]. Yet these methods remain vulnerable to adjoining-room interference, load balancing, mixed indoor-outdoor traffic, and domain shift across room types [52] [53] [54] [55].

BLE and passive Bluetooth are attractive because they can operate with low power and sometimes without dedicated infrastructure, but their failure modes are pronounced. RSSI instability under NLOS, multipath, orientation, and crowd density causes distance and localization errors; occupancy estimation also suffers when people carry multiple devices, leave devices behind, or disable Bluetooth [56] [57] [58] [59]. Context-aware calibration and filtering can improve robustness, but the evidence does not provide strong direct BLE-vs-camera or BLE-vs-Wi-Fi comparisons in dense campus crowds [60] [61] [62].

Fusion is clearly promising but under-validated. Studies combine Wi-Fi with cameras, energy, water, environmental signals, GPS, Android devices, or Bluetooth, and some use preprocessing or matching to reduce outliers and double-counting [35] [63] [64] [65]. However, no supplied study directly compares camera, Wi-Fi/CSI, BLE/RSSI, LiDAR, and fusion head-to-head in the same dense indoor-outdoor campus setting [2] [3] [4] [5] [7].

Practical implication: campuses should deploy **accurate chokepoint counters** at entrances, stairs, and sidewalks, then use **ambient wireless sensing** for broad occupancy fields elsewhere [2] [3] [35].

Confidence level: High - the accuracy/coverage tradeoff and value of fusion are consistent across the evidence [1] [2] [3] [5].

1.2 Crowd prediction and anomaly detection

Prediction capability is stronger than actuation capability in the current literature. A particularly relevant campus example used anonymized Wi-Fi AP counts with a prefix-based early-warning rule to detect lunchtime canteen buildup; it identified an Early Peak pattern by 11:28, with 0.73 coverage and 0.86 agreement, and badge days were associated with about 9% lower peak crowding [8]. This is important because it demonstrates that **emerging congestion can be detected before full crowd formation** using relatively coarse anonymous data [8].

For broader forecasting, graph-based and multimodal models show strong short-horizon performance. A campus pedestrian-flow model using STMHAGCN-KG reported large improvements over ARIMA at a 15-minute horizon, reducing RMSE and MAE by 50.72% and 61.17% respectively [10]. More generally, trajectory history plus semantic or environmental context improves performance over simpler historical baselines [66] [67].

Anomaly detection is also active, especially in vision-based safety monitoring. YOLO-family, pose-flow, 3DCNN, and related pipelines are described as near-real-time and capable of distinguishing some dangerous patterns from ordinary congregation, although the excerpts often report accuracy or AUC rather than full false-alarm tradeoffs [68] [69] [70] [71]. This matters operationally because campuses need to distinguish normal social gathering from risky congestion or unsafe behavior [44] [72].

Taken together, these findings suggest that campuses can reliably forecast **where risk is rising**, but much less evidence shows how those predictions are converted into successful route interventions [8] [10] [22].

Confidence level: High for detectability and forecasting; **Medium** for operational crowd-surge intervention because reporting on alert outcomes is limited [8] [10] [14].

1.3 Operational thresholds for emerging congestion

The literature provides threshold concepts, but not a standardized campus rulebook. Reported thresholds include 3.5 ped/m² for crowding or panic onset, roughly 6.875-6.971 ped/m² for instability, 5-6 persons/m² as a prompt-action threshold, and 0.8 m^{-2} as a behavioral shift point [11] [36] [73] [74]. Some studies use empirical or simulated tuning, such as an exit-congestion threshold with a local optimum around 0.30 and a useful band from 0.25 to 0.35, balancing congestion reduction against reassignment overhead and instability [13].

For campus operations, this implies thresholds should be **multi-factor and context-specific**: not just density, but also duration, occupancy-change rate, queue spillback risk, and schedule dependence such as class transitions or late-evening facility shutdowns [9] [36]. One campus abnormality monitor used a decision threshold and generated a real-time stampede prediction alert, showing feasibility but not broad calibration maturity [12].

Practical implication: campuses should start with rule-based thresholds, but tune them locally by building type, time block, and intervention cost [13] [14].

Confidence level: Medium - threshold concepts are well evidenced, but campus-specific validation is limited [11] [13] [14].

2. Privacy-Preserving Monitoring and Governance

2.1 Edge processing and anonymized sensing

The clearest privacy trend is **edge-first processing with metadata-only export**. Campus video analytics can keep frame filtering, inference, and face obfuscation on-device, reducing bandwidth, storage, and exposure of raw imagery [15]. A CPU-only fog architecture for university libraries used low-cost top-view cameras and lightweight processing for occupancy and flow monitoring [18]. Non-camera systems similarly aggregate radar, LiDAR, Wi-Fi, or Bluetooth measurements into region-level densities rather than identity-linked streams [49].

Wireless sensing also supports low-identifiability designs. Wi-Fi-based crowd monitoring can use probe-request messages and Bloom filters, while campus telemetry can be anonymized, limited to AP/building locations, or reduced to zone-level counts [7] [19] [75]. Metadata-only systems further reduce re-identification risk by replacing identifiers with UUIDs, rotating them every 24 hours, and applying k-anonymity filters [16].

The limitation is that anonymity is not free: utility can decline when identifiers are hidden, low-density cells are suppressed, or anonymous devices submit faulty data [76] [77].

Confidence level: High - edge and metadata-only approaches are one of the most consistent patterns in the evidence [15] [16] [18].

2.2 Privacy-preserving learning and aggregation

Federated learning, secure aggregation, differential privacy, and homomorphic encryption are repeatedly proposed to allow cross-building or cross-sensor model improvement without centralizing raw data [17] [38] [39] [78]. The tradeoff is also consistent: differential privacy can preserve useful prediction quality in some cases, but too much noise degrades accuracy; secure aggregation hides client updates but raises communication cost and impedes some anomaly-detection methods; homomorphic encryption protects aggregation but increases computational overhead and reduces training efficiency [37] [38] [78] [79] [80].

One important nuance is that not all private learning results are negative: one retrieved FL study reported accurate DP models, and another encrypted FL framework was described as lossless relative to centralized learning [78]. So the tradeoff is real, but context-dependent rather than universally prohibitive [39] [78].

Practical implication: for campus deployment, privacy-preserving learning is best reserved for **cross-building adaptation and periodic model updates**, while low-latency inference should remain local at the edge [31] [39] [81].

Confidence level: High - privacy-utility-resource tradeoffs are explicitly repeated across multiple sources [37] [38] [78].

2.3 Governance, consent, and fairness gaps

Documented governance practices include consent and transparency, opt-out rights, retention limits or deletion, access control, role-based permissions, audit logs, and data-sharing permissions [29] [82] [83] [84] [85] [86]. However, many papers frame privacy as a compliance or security issue rather than describing fully operational governance processes [87] [88].

Fairness is the clearest gap. The retrieved evidence does not show comparative fairness analysis across students, staff, visitors, or users with disabilities, even though device-carry behavior, accessibility needs, and surveillance sensitivity likely differ across those groups [43] [47] [89]. This is especially important because wireless sensing can systematically undercount people who do not carry devices or who carry them atypically [57] [59].

Confidence level: High - the governance gap, especially around fairness, is directly supported [29] [43] [90].

3. Dynamic Route Recommendation and Pedestrian Guidance

3.1 Congestion-aware route optimization

Routing research on campuses is moving from shortest-path logic to **dynamic, multi-criteria path selection**. One framework explicitly uses real-time predictive factors to reweight A* search under congestion, impediments, weather, and occupant behavior [34]. Related dynamic network-flow guidance aims for system-optimal performance by updating routing instructions with real-time information rather than merely minimizing individual path length [91]. Hybrid simulation-assignment work likewise allows pedestrians to change destination or route when congestion rises [92].

This is conceptually strong, but computational limits matter. Dynamic traffic assignment and repeated rerouting can be expensive, slow to converge, and hard to calibrate for large networks [93] [94] [95]. That makes pure network-wide optimization less practical for minute-by-minute campus operations than lighter **path-cost reweighting on a campus graph** [34] [96].

Confidence level: Medium - optimization concepts are strong, but direct campus field validation is limited [34] [92] [93].

3.2 Alternative route recommendation and user compliance

Mobile and map-based campus systems can present multiple routes, landmarks, and wayfinding cues, and some papers note that apps may help users avoid or seek crowds in real time [40] [97] [98]. Yet the crucial evidence gap is behavioral: no retrieved paper explicitly reports a campus system that gives real-time congestion-avoiding alternative routes and also measures adherence, route-choice change, or travel-time savings against static guidance [22] [24] [99].

This means campuses can already build the interface layer, but they should treat the **human response model** as an open problem. The evidence supports usability and acceptance of AR and mobile wayfinding more generally, including good SUS, TAM, UEQ, and task-completion results in student-heavy studies [100] [101] [102] [103]. What it does not yet show is whether users follow live congestion recommendations when those recommendations conflict with habit or shortest distance [22].

Practical implication: route guidance should be tested with A/B comparisons among static maps, live congestion maps, and explicit reroute nudges [22] [103].

Confidence level: High on the existence of the evidence gap; **Low-Medium** on expected operational gains because they are not directly measured [22] [24].

3.3 Accessibility-aware campus navigation

Accessibility is one of the better-developed routing themes. Campus routing studies encode slope, paving quality, obstacles, stairs, elevators, entrances, and indoor-outdoor transitions through multi-criteria ranking or graph refinement [21] [46] [47] [104]. AHP-based studies rank ramp slope and paving as highly important for accessible route selection [46]. Indoor-outdoor campus routing requires fine-grained level data plus vertical connectors and building entrances, with wheelchair use explicitly considered [21]. Preference-based pedestrian routing also shows that users may avoid crowded areas, cafeterias, lecture halls, unsafe areas, or escalators depending on context [23].

Taken together, these findings suggest that congestion-aware routing should not be added as a separate layer after accessibility; instead, both should be encoded jointly in route cost functions [21] [23] [42].

Confidence level: High - accessibility-aware path logic is clearly documented, even if congestion-aware field evaluation is thin [21] [46] [47].

4. Deployment, Alerts, and Scalable Smart-Campus Operations

4.1 Automated alerts and incident workflows

Campus and adjacent smart-safety systems commonly generate real-time alerts from occupancy, unsafe behavior, unauthorized access, or anomaly detection [26] [28] [105]. Some systems include escalation logic: one campus workflow sent repeated reminders to managers until feedback was provided, while another worker-safety system escalated from alert to management notification, equipment shutdown, and emergency procedures [26] [27]. Security-oriented systems can notify guards or responsible staff directly [28] [106].

For crowd management specifically, the evidence supports **automated detection plus operator notification**, but not yet richly documented override procedures or event-resolution playbooks [25] [26]. This suggests a need for **human-in-the-loop dashboards** that show zone status, forecast risk, recommended interventions, and audit traces [25] [48].

Confidence level: Medium - alerting is well supported, but detailed campus crowd-response workflows are not [26] [27] [28].

4.2 Scalable and interoperable architectures

Scalable architectures typically use layered integration: data acquisition, edge computing, cloud services, application logic, and dashboards [29] [31]. Interoperability is framed around unifying heterogeneous sensors, indoor/outdoor protocols, and campus applications, with some work emphasizing standards-style separation of sensing and tasking services [29] [30]. Multi-building Wi-Fi learning systems such as MetaMobi and ACroCI directly address cross-building mobility learning and adaptation to altered AP infrastructure [107] [108].

However, scalability evidence remains brittle. Many systems are ad hoc, single-use-case, or single-zone, and mixed indoor-outdoor validation is inconsistent across duration, metrics, and context [32] [33] [109]. This means architectural plausibility is ahead of proven operational generalization [110] [111].

Confidence level: High - the recurring architecture pattern is clear, and so is the weakness of the deployment evidence [29] [31] [32].

4.3 Evaluation metrics and unresolved gaps

Evaluation is uneven across layers. Sensing papers commonly report accuracy, precision, recall, F1, RMSE, or processing speed [112] [113] [114]. Forecasting papers emphasize regression metrics such as RMSE and MAE [10] [115]. Safety-alert work is the richest operationally, including warning lead time, false alarms per hour or shift, missed-event rate, and event-level detection measures [14]. But rerouting studies rarely report dedicated metrics, and the supplied evidence does not show user-compliance measures for dynamic crowd guidance [22] [41].

This points to four major research gaps:

1. **No integrated benchmark** across sensing modalities in the same campus environment [6] [7].
2. **Limited operational alert evaluation** for campus congestion prediction [8] [14].
3. **Almost no field evidence on reroute adherence or bottleneck reduction** [22] [24].
4. **Weak governance and fairness evaluation** across user groups [29] [43].

Confidence level: High - these gaps are explicit and repeated across the corpus [6] [14] [22] [43].

5. Recommended Integration Blueprint for Scalable Smart-Campus Solutions

A synthesis of the evidence supports the following implementable design:

5.1 Sensing layer

- Deploy **camera or LiDAR sentinel counters** at chokepoints, entrances, stairs, elevators, and major sidewalks for accurate flow/count measurement [2] [3].
- Use **Wi-Fi/CSI and BLE ambient sensing** for room-, corridor-, and zone-level occupancy fields where visual coverage is incomplete or privacy-sensitive [5] [59].
- Apply filtering for pass-by traffic, adjacent-room bleed, inactive devices, and multi-device bias [52] [116].

5.2 Prediction layer

- Convert sensor feeds into a **zone graph** with occupancy, inflow, outflow, and dwell proxies [8] [10].
- Run short-horizon models for 5-15 minute risk forecasting using AP trends, semantic trajectories, and graph-based temporal attention [8] [10] [107].
- Add anomaly detectors for unsafe behavior or abnormal buildup where appropriate [44] [68].

5.3 Decision and alert layer

- Combine learned forecasts with **rule-based thresholds** on density, occupancy-change rate, and persistence [11] [36].
- Trigger tiered alerts to dashboards, staff, or user channels based on severity and duration [25] [28].
- Maintain audit logs and role-based permissions for accountability [25] [86].

5.4 Guidance layer

- Reweight route costs by predicted congestion, accessibility, indoor-outdoor transitions, shelter, and restrictions [21] [23] [34].
- Deliver guidance through **apps, AR, signage, and operator dashboards**, allowing different interventions for students, visitors, and staff [40] [48] [117].
- Keep a manual override path for exceptional cases because automated intervention logic is under-documented [26] [48].

5.5 Privacy and governance layer

- Prefer **on-device inference**, **metadata-only export**, **rotating identifiers**, and **federated model updates** [15] [16] [17].
- Document retention, opt-out, access roles, audit logs, and data-sharing rules from the start [82] [85] [86].
- Add explicit fairness review for device-carry bias and accessibility impacts [43] [47] [59].

Inferred: A practical first deployment should focus on a few high-risk transition windows such as lunch peaks and class changes, because these periods already have stronger evidence for detectable buildup and thresholded intervention than all-day full-campus optimization [8] [9]. **Confidence level: Low-Medium** - this is a synthesis-based deployment inference rather than a directly validated end-to-end result [8] [9].

Conclusion

The evidence indicates that integrated smart-campus crowd management is **technically feasible but operationally incomplete** [2] [8] [29]. Real-time sensing works best when accurate chokepoint counters are fused with broader wireless occupancy fields [3] [5]. Short-horizon prediction from Wi-Fi and graph-based models can detect emerging congestion before full crowd formation [8] [10]. Privacy-preserving designs are increasingly edge-first and metadata-centric, reducing reliance on centralized raw personal data [15] [16]. Routing research is mature enough to encode congestion and accessibility jointly, but field evidence on live rerouting benefits and user compliance is notably thin [21] [22] [34]. Operationally, dashboards, alerts, and layered architectures exist, yet standardized alert metrics, override workflows, fairness audits, and long-term multi-building validation remain underdeveloped [14] [25] [32] [43].

The strongest opportunity is therefore a **scalable edge-first, multimodal, privacy-aware campus platform** that closes the loop from sensing to prediction to guidance, while measuring not just model accuracy but also lead time, false alarms, adherence, accessibility outcomes, and perceived fairness [14] [15] [22] [47].

Overall conclusion confidence: High for the architectural synthesis and identified gaps, **Medium** for expected real-world impact until closed-loop campus trials are performed [2] [8] [22] [29].

References

1. Anand, P., Cheong, D., Sekhar, C., (...), Kondepudi, S. (2019) Energy saving estimation for plug and lighting load using occupancy analysis. *Renewable Energy*. https://doi.org/10.1016/j.renene.2019.05.089
2. Sun, K., Wang, X., Xing, T., (...), Zhao, Q. (2024) High-accuracy occupancy counting at crowded entrances for smart buildings. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2024.114509
3. Lesani, A., Nateghinia, E., Miranda-Moreno, L.F. (2020) Development and evaluation of a real-time pedestrian counting system for high-volume conditions based on 2D LiDAR. *Transportation Research Part C: Emerging Technologies*. https://doi.org/10.1016/j.trc.2020.01.018
4. Anand, P., Deb, C., Yan, K., (...), Sekhar, C. (2021) Occupancy-based energy consumption modelling using machine learning algorithms for institutional buildings. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2021.111478
5. Nweye, K., Nagy, Z. (2022) MARTINI: Smart meter driven estimation of HVAC schedules and energy savings based on Wi-Fi sensing and clustering. *Applied Energy*. https://doi.org/10.1016/j.apenergy.2022.118980
6. Mammadov, S., Kucukkulahli, E. (2025) A User-Centric Smart Library System: IoT-Driven Environmental Monitoring and ML-Based Optimization with Future Fog–Cloud Architecture. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app15073792
7. Amiri, A.J., Mahmoodi, K., Tootchi, A., Razban, A. (2025) A review of occupancy detection techniques for HVAC control: Advances and practical challenges. *Journal of Building Engineering*. https://doi.org/10.1016/j.jobe.2025.113962
8. Variskhanov, A., Arreeras, T. (2026) Early Peak Badges from Wi-Fi Telemetry: A Field Feasibility Study of Lunchtime Crowd Management on a Smart Campus. *Urban Science*. https://doi.org/10.3390/urbansci10010029

9. Wang, Mingzhu, Wong, Peter Kok-Yiu, Cheng, Jack C. P. (2025) A BIM-GIS Framework Integrated with CCTV Analytics for Urban Walkability Assessment. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25123637
10. https://www.sciencedirect.com/science/article/pii/S1573749725006071
11. Qinghua, M., Songzhuo, H., Li, L., Wan, X. (2026) A group evacuation simulation model under bidirectional flow collision and stampede scenarios. *Physica A: Statistical Mechanics and its Applications*. https://doi.org/10.1016/j.physa.2026.131371
12. Vinodini Ramesh, M.V., Anjitha, A., Prabha, R. (2014) Context aware ad hoc network for mitigation of crowd disasters. *Ad Hoc Networks*. https://doi.org/10.1016/j.adhoc.2013.02.006
13. Zhou, L., Wei, Q., Yang, Y., (...), Li, K. (2026) An adaptive large neighborhood search with dynamic exit reassignment for guided multi-story hospital fire evacuation. *Information Sciences*. https://doi.org/10.1016/j.ins.2026.123475
14. Yu, K., Jiang, W. (2026) A process-safety-oriented BiTCN-BiGRU-Attention prediction-to-DBN evidence framework for methane over-limit warning and safety-barrier reasoning in underground coal mines. *Process Safety and Environmental Protection*. https://doi.org/10.1016/j.psep.2026.109559
15. Amo-Filva, D., Fonseca, D., García-Peñalvo, F.J., (...), Godoy, G. (2024) Exploring the landscape of learning analytics privacy in fog and edge computing: A systematic literature review. *Computers in Human Behavior*. https://doi.org/10.1016/j.chb.2024.108303
16. Magyar, M., Ala-Hulkko, T., Antikainen, H., (...), Kotavaara, O. (2025) Utilizing mobile phone tracking data to estimate Intra-City modal mobility: A study on active mobility in two Finnish City regions. *Journal of Transport Geography*. https://doi.org/10.1016/j.jtrangeo.2025.104326
17. Houichi, M., Jaïdi, F., Bouhoula, A. (2026) A trust-aware federated intrusion detection framework for privacy-preserving smart city IoT networks. *Computer Networks*. https://doi.org/10.1016/j.comnet.2026.112616
18. Martínez-Sala, A.S., Hernando-Cánovas, L., Sanchez-Aarnoutse, J.C., Alcaraz, J.J. (2025) Resource-efficient fog computing vision system for occupancy monitoring: A real-world deployment in university libraries. *Internet of Things (The Netherlands)*. https://doi.org/10.1016/j.iot.2025.101748
19. Petros, B.A., Paull, J.S., Tomkins-Tinch, C.H., (...), Sabeti., P.C. (2022) Multimodal surveillance of SARS-CoV-2 at a university enables development of a robust outbreak response framework. *Med*. https://doi.org/10.1016/j.medj.2022.09.003

20. Vinod, N., Divija Reddy, K., Hari Priya, K., (...), Pillai, K.N. (2026) A Graph-Based Smart Campus Navigation System Using A∗ Search and Multi-Criteria Route Evaluation. *Proceedings of 3rd International Conference on Electronics, Computing, Communication and Control Technology: AI-Driven Smart Electronics and Next-Generation Communication Technologies, ICECCC 2026*. https://doi.org/10.1109/ICECCC70334.2026.11633005

21. Li, R., Storandt, S., Müller, U., Weber, D. (2021) Barrier-Free Pedestrian Routing with Contraction Hierarchies. *GIS: Proceedings of the ACM International Symposium on Advances in Geographic Information Systems*. https://doi.org/10.1145/3474717.3486797

22. Angelelli, E., Arşik, I., Morandi, V., (...), Speranza, M.G. (2016) Proactive route guidance to avoid congestion. *Transportation Research Part B: Methodological*. https://doi.org/10.1016/j.trb.2016.08.015

23. Müller, M., Ohm, C., Schwappach, F., Ludwig, B. (2017) The path of least resistance: Calculating preference adapted routes for pedestrian navigation. *KI - Kunstliche Intelligenz*. https://doi.org/10.1007/s13218-016-0472-6

24. Fries, Ryan N., Dunning, Anne E., Chowdhury, Mashrur A. (2011) University Traveler Value of Potential Real-Time Transit Information. *Journal of Public Transportation*. https://doi.org/10.5038/2375-0901.14.2.2

25. Min-Allah, N., Alrashed, S. (2020) Smart campus—A sketch. *Sustainable Cities and Society*. https://doi.org/10.1016/j.scs.2020.102231

26. Chen, Huihua, Li, Yaoyu, Wen, Huanxi, Hu, Xiaodong (2023) *Frontiers in Public Health*. https://doi.org/10.3389/fpubh.2023.1225478

27. Williams, S., Fiumara, K., Kachalia, A., Desai, S. (2020) Closing the Loop with Ambulatory Staff on Safety Reports. *Joint Commission Journal on Quality and Patient Safety*. https://doi.org/10.1016/j.jcjq.2019.09.009

28. Villegas-Ch, William, García-Ortiz, Joselin (2023) Authentication, access, and monitoring system for critical areas with the use of artificial intelligence integrated into perimeter security in a data center. *Frontiers in Big Data*. https://doi.org/10.3389/fdata.2023.1200390

29. https://www.sciencedirect.com/science/article/pii/S1572834X20004840

30. Alrashed, S. (2020) Key performance indicators for Smart Campus and Microgrid. *Sustainable Cities and Society*. https://doi.org/10.1016/j.scs.2020.102264

31. Siddiqa, A., Khan, W.Z., Alkinani, M.H., (...), Khan, M.K. (2024) Edge-assisted federated learning framework for smart crowd management. *Internet of Things (Netherlands)*. https://doi.org/10.1016/j.iot.2024.101253

32. Huertas Celdrán, Alberto, Ruipérez-Valiente, José A., García Clemente, Félix J., (...), Martínez Pérez, Gregorio (2020) A Scalable Architecture for the Dynamic Deployment of Multimodal Learning Analytics Applications in Smart Classrooms. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s20102923
33. Ammar, M.A., Wibowo, M.A., Nurhuda, I., Budihardjo, M.A. (2026) Sustainable energy and infrastructure technologies in university campuses: A cross-technology comparative analysis. *Next Energy*. https://doi.org/10.1016/j.nxener.2026.100964
34. David, S., Pavithra, M.H., Shreen Fathima, S. (2025) Machine Learning-Guided A* Framework for Adaptive Pathfinding in Smart Campus Systems. *Proceedings - IEEE 10th International Conference on Smart Structures and Systems, ICSSS 2025*. https://doi.org/10.1109/ICSSS66939.2025.11346172
35. https://www.sciencedirect.com/science/article/pii/S1572834X24003137
36. Wang, J., Zhu, Y., Shen, L., (...), Weng, W. (2026) Empirical study on spatial characteristics of pedestrian groups in a curved passage. *Journal of Building Engineering*. https://doi.org/10.1016/j.jobe.2026.116117
37. https://www.sciencedirect.com/science/article/pii/S1424822026039968
38. Yuan, Y., Rao, H., Liu, S., (...), Hu, Y. (2026) FedOut: Federated learning with outsourced homomorphic encryption for resource-constrained IoT devices. *Advanced Engineering Informatics*. https://doi.org/10.1016/j.aei.2026.105162
39. Alabdulatif, A. (2025) GuardianAI: Privacy-preserving federated anomaly detection with differential privacy. *Array*. https://doi.org/10.1016/j.array.2025.100381
40. Traunmueller, M.W., Johnson, N., Malik, A., Kontokosta, C.E. (2018) Digital footprints: Using WiFi probe and locational data to analyze human mobility trajectories in cities. *Computers, Environment and Urban Systems*. https://doi.org/10.1016/j.compenvurbsys.2018.07.006
41. https://www.sciencedirect.com/science/article/pii/S2730723926000370
42. Arenghi, A., Fogli, D., Belometti, S., (...), Plebani, N. (2018) UniBs4All: A mobile application for accessible wayfinding and navigation in an urban university campus. *ACM International Conference Proceeding Series*. https://doi.org/10.1145/3284869.3284900
43. Cheong, Pauline Hope, Nyaupane, Pratik (2022) Smart campus communication, Internet of Things, and data governance: Understanding student tensions and imaginaries. *Big Data & Society*. https://doi.org/10.1177/20539517221092656
44. Wu, Y., Feng, S., Wu, Y., (...), Wu, C. (2025) Masked region disparity-based unsafe behavior detection via campus monitoring device images. *Knowledge-Based Systems*. https://doi.org/10.1016/j.knosys.2025.113051

45. https://www.sciencedirect.com/science/article/pii/S2198605322003005

46. Kocaaslan, H., Pouya, S., Sönmez, S. (2024) Finding the optimal route for people with mobility impairments: A case study of the İnönü University campus. *Urbani Izziv*. https://doi.org/10.5379/urbani-izziv-en-2024-35-01-03

47. Minetto, R., Kozievitch, N.P., da Silva, R.D., (...), de Santi, J. (2016) Shortcut suggestion based on collaborative user feedback for suitable wheelchair route planning. *IEEE Conference on Intelligent Transportation Systems, Proceedings, ITSC*. https://doi.org/10.1109/ITSC.2016.7795938

48. https://www.sciencedirect.com/science/article/pii/S1573754326006056

49. Gayathri, S.V.L., Harshitha, T., Kriyasri, H., Balasundaram, A. (2025) Enhancing crowd management through behaviourally informed synthetic datasets and predictive deep learning models. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2025.108210

50. Rusca, R., Gasco, D., Casetti, C., Giaccone, P. (2024) Privacy-preserving WiFi fingerprint-based people counting for crowd management. *Computer Communications*. https://doi.org/10.1016/j.comcom.2024.07.010

51. Lavanya, R., Murukesh, C., Shanker, N.R. (2023) Microclimatic HVAC system for nano painted rooms using PSO based occupancy regression controller. *Energy*. https://doi.org/10.1016/j.energy.2023.127828

52. Ren, J., Zhou, X., Jin, X., (...), Shi, X. (2024) A systematic review of occupancy pattern in urban building energy modeling: From urban to building-scale. *Journal of Building Engineering*. https://doi.org/10.1016/j.jobe.2024.110307

53. Mohottige, I.P., Moors, T. (2018) Estimating Room Occupancy in a Smart Campus using WiFi Soft Sensors. *Proceedings - Conference on Local Computer Networks, LCN*. https://doi.org/10.1109/LCN.2018.8638098

54. Mohottige, I.P., Sutjarittham, T., Raju, N., (...), Sivaraman, V. (2018) Role of Campus WiFi Infrastructure for Occupancy Monitoring in a Large University. *2018 IEEE 9th International Conference on Information and Automation for Sustainability, ICIAfS 2018*. https://doi.org/10.1109/ICIAFS.2018.8913341

55. Jung, M., Lee, J., Kim, D., (...), Kim, T. (2025) Implementing Wi-Fi CSI-based room-level occupancy Estimation: an experimental study in multi-zone residential environments. *Journal of Building Engineering*. https://doi.org/10.1016/j.jobe.2025.113155

56. https://www.sciencedirect.com/science/article/pii/S2661890726002656

57. Azimi, S., O'Brien, W. (2022) Fit-for-purpose: Measuring occupancy to support commercial building operations: A review. *Building and Environment*. https://doi.org/10.1016/j.buildenv.2022.108767

58. Flueratoru, L., Shubina, V., Niculescu, D., Lohan, E.S. (2022) On the High Fluctuations of Received Signal Strength Measurements with BLE Signals for Contact Tracing and Proximity Detection. *IEEE Sensors Journal*. https://doi.org/10.1109/JSEN.2021.3095710

59. Datla, V.S.V., Aiuti, A., Bisante, A., (...), Panizzi, E. (2025) Detecting Human Presence via Smartphone BLE Beaconing: Preliminary Investigations. *Proceedings of MUM 2025 - The 24th International Conference on Mobile and Ubiquitous Multimedia*. https://doi.org/10.1145/3771882.3773958

60. Datla, V.S.V., Aiuti, A., Bisante, A., (...), Panizzi, E. (2026) Infrastructure-Free Indoor Occupancy Estimation via Passive BLE Scanning. *Proceedings of the ACM on Human-Computer Interaction*. https://doi.org/10.1145/3816778

61. Datla, V.S.V., Bisante, A., Trasciatti, G., (...), Panizzi, E. (2026) Demonstrating Infrastructure-Free Indoor Occupancy Visualization using Passive BLE Sensing. *Proceedings of the 2026 International Conference on Advanced Visual Interfaces, AVI 2026*. https://doi.org/10.1145/3811427.3811507

62. Mahmud, S., Bensaali, F., Chowdhury, M.E.H., Houchati, M. (2025) Multimodal feature fusion and ensemble learning for non-intrusive occupancy monitoring using smart meters. *Building and Environment*. https://doi.org/10.1016/j.buildenv.2025.112635

63. Ravi, A., Misra, A. (2020) Robust, Fine-Grained Occupancy Estimation via Combined Camera WiFi Indoor Localization. *Proceedings - 2020 IEEE 17th International Conference on Mobile Ad Hoc and Smart Systems, MASS 2020*. https://doi.org/10.1109/MASS50613.2020.00074

64. Himeur, Y., Alsalemi, A., Al-Kababji, A., (...), Amira, A. (2020) Data fusion strategies for energy efficiency in buildings: Overview, challenges and novel orientations. *Information Fusion*. https://doi.org/10.1016/j.inffus.2020.07.003

65. Markus, A.A., Hobson, B.W., Gunay, H.B., Bucking, S. (2021) A framework for a multi-source, data-driven building energy management toolkit. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2021.111255

66. Vakrinou, K., Mantouka, E.G., Vlahogianni, E.I. (2026) Leveraging multisource data for off-street parking occupancy prediction: a twofold analysis. *International Journal of Transportation Science and Technology*. https://doi.org/10.1016/j.ijtst.2025.02.005

67. Han, Yuting, Li, Shuyu, Tan, Yunfei (2026) Attention-Enhanced Pedestrian Trajectory Prediction via Compressed Point Cloud Representation. *Journal of Imaging*. https://doi.org/10.3390/jimaging12070305

68. Pranathi, M.D., Satyanarayana, M., Nethala, T.R., Kumar, K.R. (2026) CAMPUS-INTENT:A Hybrid Deep Learning Framework Integrated with YOLOv8 for Intelligent Crowd Intention Recognition in Educational Campuses. *2026 1st IEEE International Conference on Intelligent Technologies for a Sustainable and Inclusive Future, ICITSIF 2026*. https://doi.org/10.1109/ICITSIF69060.2026.11608977

69. Mohamed, S.A.E.-M., Askr, H., Hassanien, A.E. (2026) A generative adversarial network framework for intelligent crowd anomaly detection in metaverse airport surveillance. *Applied Soft Computing*. https://doi.org/10.1016/j.asoc.2026.115849

70. https://www.sciencedirect.com/science/article/pii/S1573772121006888

71. https://www.sciencedirect.com/science/article/pii/S1573772123000510

72. Mohammadi, S., Kiani, H.K., Perina, A., Murino, V. (2017) Physics-Inspired Models for Detecting Abnormal Behaviors in Crowded Scenes. *Group and Crowd Behavior for Computer Vision*. https://doi.org/10.1016/B978-0-12-809276-7.00013-8

73. Du, X., Chen, Y., Bouferguène, A., Al-Hussein, M. (2020) An agent-based simulation framework for analysing fall risk among older adults in the evacuation procedures. *Safety Science*. https://doi.org/10.1016/j.ssci.2020.104790

74. Lu, Hongcheng, Wang, Siming, Ye, Ran, (...), Wang, Yan (2025) Research on critical criteria for crowd instability based on system entropy. *Simulation Modelling Practice and Theory*. https://doi.org/10.1016/j.simpat.2025.103213

75. Liu, Xingyu, Meng, Xin, Duan, Hancong, (...), Wang, Min (2025) A Survey on Secure WiFi Sensing Technology: Attacks and Defenses. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25061913

76. Reinhardt, D., Reinhardt, A., Kanhere, S.S., Hollick, M. (2011) A survey on privacy in mobile participatory sensing applications. *Journal of Systems and Software*. https://doi.org/10.1016/j.jss.2011.06.073

77. Wang, X., Liu, Z., Tian, X., (...), Wang, X. (2017) Incentivizing Crowdsensing With Location-Privacy Preserving. *IEEE Transactions on Wireless Communications*. https://doi.org/10.1109/TWC.2017.2734758

78. Michalakopoulos, V., Sarantinopoulos, E., Sarmas, E., Marinakis, V. (2024) Empowering federated learning techniques for privacy-preserving PV forecasting. *Energy Reports*. https://doi.org/10.1016/j.egyr.2024.08.033

79. Niu, S., Zhou, X., Wang, N., (...), Chen, L. (2025) Privacy-preserving and efficient verifiable federated learning scheme based on proxy re-signature. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2025.128422

80. Moshawrab, Mohammad, Adda, Mehdi, Bouzouane, Abdenour, (...), Raad, Ali (2023) Reviewing Federated Machine Learning and Its Use in Diseases Prediction. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s23042112

81. Siddiqa, A., Khan, W.Z., Alkinani, M.H., (...), Khan, M.K. (2024) Edge-assisted federated learning framework for smart crowd management. *Internet of Things (Netherlands)*. https://doi.org/10.1016/j.iot.2024.101253

82. Yoon, Sungwook, Kim, Byungmun (2025) Multi-Scale Temporal Fusion Network for Real-Time Multimodal Emotion Recognition in IoT Environments. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25165066

83. Zeng, S., Chen, X., Su, D., Gong, H. (2025) Multi-source data-driven intelligent analysis and decision optimization for high-density pedestrian flows in urban public spaces. *Automation in Construction*. https://doi.org/10.1016/j.autcon.2025.106367

84. Van de Glind, Geurt, Crilly, Julia, Galenkamp, Niek, (...), Ham, Wietske H W (2024) Defining the concept of mental dysregulation in patients requiring ambulance and/or emergency department care: protocol for a Delphi consensus study. *BMJ Open*. https://doi.org/10.1136/bmjopen-2023-077666

85. van Leeuwen, A., Goudriaan, M., Aksu, Ü. (2024) How to responsibly deploy a predictive modelling dashboard for study advisors? A use case illustrating various stakeholder perspectives. *Computers and Education: Artificial Intelligence*. https://doi.org/10.1016/j.caeai.2024.100304

86. Divadari, Satyavathi (2025) Cloud security for smart sensor network. *Sensor Networks for Smart Hospitals*. https://doi.org/10.1016/B978-0-443-36370-2.00023-2

87. Stanciu, V.-D., Van Steen, M., Dobre, C., Peter, A. (2023) Privacy-friendly statistical counting for pedestrian dynamics. *Computer Communications*. https://doi.org/10.1016/j.comcom.2023.09.009

88. Monti, Lorenzo, Tse, Rita, Tang, Su-Kit, (...), Salomoni, Paola (2022) Edge-Based Transfer Learning for Classroom Occupancy Detection in a Smart Campus Context. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s22103692

89. Toutouh, Jamal, Alba, Enrique (2022) A Low Cost IoT Cyber-Physical System for Vehicle and Pedestrian Tracking in a Smart Campus. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s22176585

90. Sarhan, A.Y. (2026) A privacy-aware Artificial Intelligence framework for crowd monitoring and dispatching using wearable Internet of Things. *Engineering Applications of Artificial Intelligence*. https://doi.org/10.1016/j.engappai.2026.115516

91. Feng, L., Miller-Hooks, E. (2014) A network optimization-based approach for crowd management in large public gatherings. *Transportation Research Part C: Emerging Technologies*. https://doi.org/10.1016/j.trc.2014.01.017

92. Abdelghany, A., Abdelghany, K., Mahmassani, H. (2016) A hybrid simulation-assignment modeling framework for crowd dynamics in large-scale pedestrian facilities. *Transportation Research Part A: Policy and Practice*. https://doi.org/10.1016/j.tra.2016.02.011

93. Patwary, A.U.Z., Huang, W., Lo, H.K. (2021) A link-to-link segment based metamodel for dynamic network loading. *Transportation Research Part C: Emerging Technologies*. https://doi.org/10.1016/j.trc.2021.103286

94. Aghamohammadi, R., Laval, J.A. (2020) Dynamic traffic assignment using the macroscopic fundamental diagram: A Review of vehicular and pedestrian flow models. *Transportation Research Part B: Methodological*. https://doi.org/10.1016/j.trb.2018.10.017

95. Park, K., Kim, W. (2001) A systolic parallel simulation system for dynamic traffic assignment: SPSS-DTA. *Expert Systems with Applications*. https://doi.org/10.1016/S0957-4174(01)00041-0

96. Rickert, M., Nagel, K. (2001) Dynamic traffic assignment on parallel computers in TRANSIMS. *Future Generation Computer Systems*. https://doi.org/10.1016/S0167-739X(00)00032-7

97. Ohm, C., Bienk, S., Kattenbeck, M., (...), Müller, M. (2016) Towards interfaces of mobile pedestrian navigation systems adapted to the user's orientation skills. *Pervasive and Mobile Computing*. https://doi.org/10.1016/j.pmcj.2015.10.006

98. Hosseini, R., Lim, S., Tong, D., (...), Seyedabrishami, S. (2024) A specialized inclusive road dataset with elevation profiles for realistic pedestrian navigation using open geospatial data and deep learning. *Computers, Environment and Urban Systems*. https://doi.org/10.1016/j.compenvurbsys.2024.102199

99. Namakula, H., Bukenya, I., Tusubira, J., (...), Bagampadde, U. (2026) Complementarity between stated and revealed preferences in pedestrian route choice: Evidence from a university campus living lab in Kampala. *Scientific African*. https://doi.org/10.1016/j.sciaf.2026.e03434

100. Marian-Vladut, T., Paul, P., Octavian, T.C. (2025) Cross-Context Evaluation of an Indoor–Outdoor AR Navigation System in a University Campus Environment. *International Journal of Advanced Computer Science and Applications*. https://doi.org/10.14569/IJACSA.2025.0160703

101. Arifitama, B., Syahputra, A. (2025) User Experience Analysis of a Mobile Augmented Reality Campus Visualization Using User Experience Questionnaire (UEQ) Approach. *Proceedings - 2025 International Seminar on Application for Technology of Information and Communication: Advancing Electrical Engineering and Informatics Through Quantum Computing and Intelligence Technologies, iSemantic 2025*. https://doi.org/10.1109/ISemantic67418.2025.11292054

102. Ramírez-Galván, E., Benavides-Álvarez, C., Aviles-Cruz, C., (...), Serrano-Talamantes, J.F. (2025) Markerless AR Navigation for Smart Campuses: Lightweight Machine Learning for Infrastructure-Free Wayfinding. *Electronics (Switzerland)*. https://doi.org/10.3390/electronics14244834

103. Ma, J., Zhang, Y. (2025) Augmented Reality for Campus Wayfinding: Enhancing Navigation Efficiency and Student Social Engagement — A Case Study of Leeds University Union. *Visible Language*. https://doi.org/10.34314/jk9zgk40

104. Zastudil, C., Stewart, M., Sprague, N., Brady, E. (2020) Campuspartner: An assistive technology for mobility impaired pedestrians. *Proceedings of the International ACM SIGGROUP Conference on Supporting Group Work*. https://doi.org/10.1145/3323994.3369893

105. Haghani, M., Coughlan, M., Crabb, B., (...), Wilson, A. (2023) A roadmap for the future of crowd safety research and practice: Introducing the Swiss Cheese Model of Crowd Safety and the imperative of a Vision Zero target. *Safety Science*. https://doi.org/10.1016/j.ssci.2023.106292

106. Bouma, Henri, Villani, Maria Luisa, van Rooijen, Arthur, (...), De Dominicis, Luigi (2022) An Integrated Fusion Engine for Early Threat Detection Demonstrated in Public-Space Trials. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s23010440

107. Yang, X., He, S., Tabatabaie, M., Wang, B. (2022) Towards Dynamic Crowd Mobility Learning and Meta Model Updates for A Smart Connected Campus. *International Conference on Embedded Wireless Systems and Networks*. https://www.scopus.com/pages/publications/85141862051

108. He, S., Wang, B., Shin, K.G., Tabatabaie, M. (2022) Cross-zone and extreme-aware mobility learning of crowd interactions with built environments. *BuildSys 2022 - Proceedings of the 2022 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*. https://doi.org/10.1145/3563357.3564065

109. https://www.sciencedirect.com/science/article/pii/S3004926126003751

110. Rodríguez-Sánchez, C., Gaitan, N.C., de Mera Sánchez, P.D., (...), Melero, J.A. (2026) Real-world validation of an IoT–BIM/AIM research testbed for building energy and indoor environmental quality monitoring. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2026.118121

111. Akdağ, F., Künyeli, F.B. (2025) Reframing the Campus: A Systematic Scientometric Review of Global Research on University Campus Design. *Architecture and Urban Planning*. https://doi.org/10.2478/aup-2025-0018

112. Al-Sa'd, Mohammad, Kiranyaz, Serkan, Ahmad, Iftikhar, (...), Gabbouj, Moncef (2022) A Social Distance Estimation and Crowd Monitoring System for Surveillance Cameras. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s22020418

113. https://www.sciencedirect.com/science/article/pii/S1433305823000229

114. Yan, Z., Liu, X., Li, G., (...), Huang, Q. (2026) Compactness driven Co-learning for crowd counting and localization. *Pattern Recognition*. https://doi.org/10.1016/j.patcog.2026.113638

115. Huang, T., Xu, Z., Ou, X., (...), Geng, J. (2025) Multi-scale convolutional attention knowledge tracing from accumulative and structural evolution perspectives. *Applied Soft Computing*. https://doi.org/10.1016/j.asoc.2025.113601

116. Dorokhova, M., Ballif, C., Wyrsch, N. (2020) Rule-based scheduling of air conditioning using occupancy forecasting. *Energy and AI*. https://doi.org/10.1016/j.egyai.2020.100022

117. Romli, R., Yew, C.C., Ghazali, N.H., (...), Hasnan, R. (2024) AR@Campus: Augmented Reality (AR) for Indoor Positioning and Navigation Apps. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0213935